

AKADEMIKA

ACL-FIL

**FILTER- NOISE-AUDIO
AMPLIFIER KIT**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ANALOG COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook, as they provide you with important points on safety during the installation, use and maintenance. Keep this handbook at hand for any further help.

Set all accessories in order on unpacking, and check integrity with respect to its checklist. Check the Kit for any visible damage.

Before connecting the power supply to the equipment, be sure patch cords are connected correctly, jumpers are correct as per experimental position.

In **ACL-FIL** kit, signals are to be monitored with an oscilloscope, the scope input channel should be a.c. coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

These kits must be employed only for the use for which it has been conceived, i.e. as educational kits, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of fault or malfunctioning of the trainers, turn off the power supply and do not tamper the kits. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit /system has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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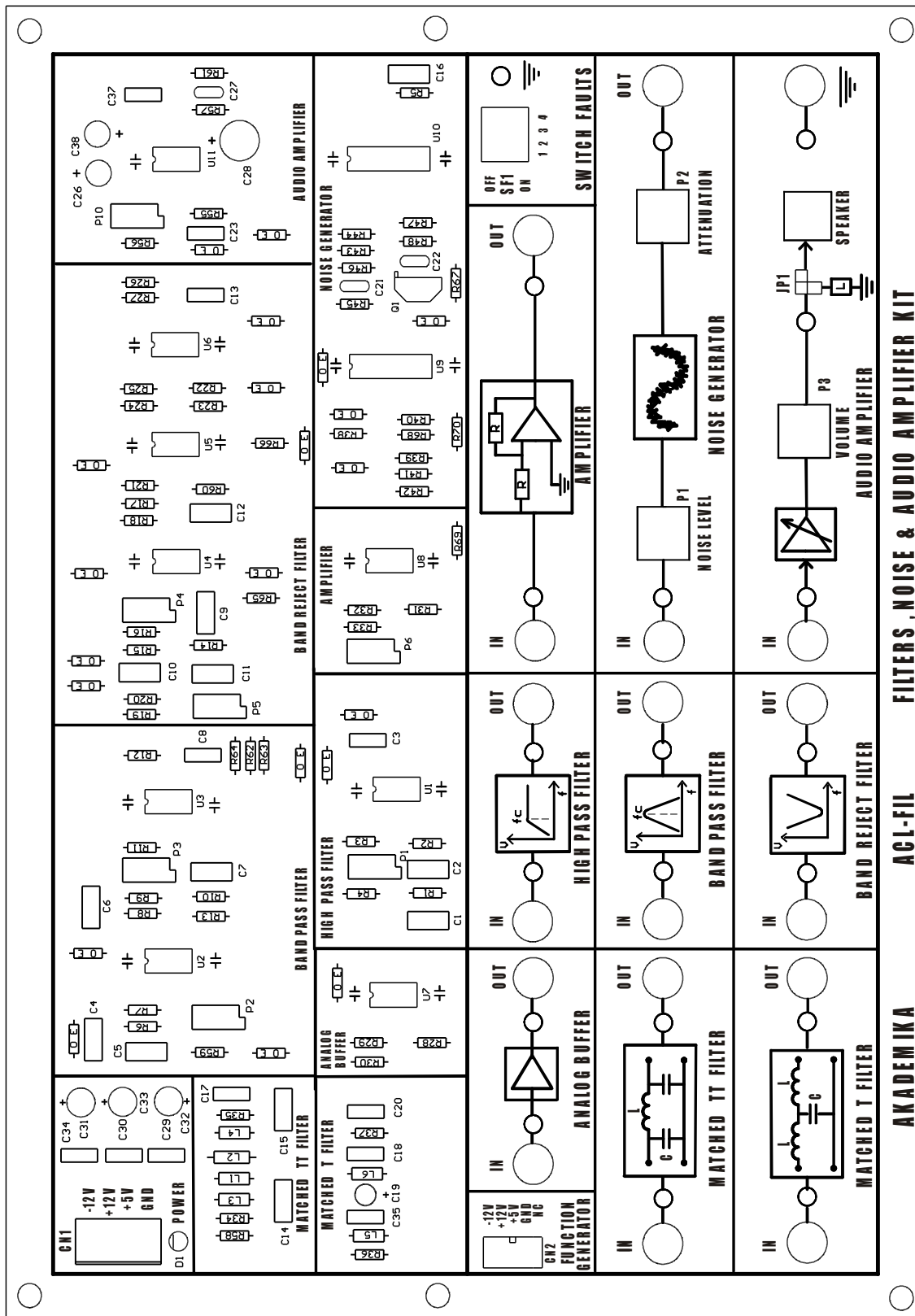
TECHNICAL SPECIFICATIONS

Noise Generator	: Output Amplitude Adjustable from 0 to 4Vp-p
Signal Attenuation Network	: Adjustable from 0 to the maximum of Input value
Audio Amplifier with Loudspeaker	: Signal + Noise Adder Stage Output Power: 0.5W. Speaker : 8 Ohms, 0.3W.
Filter	: 4 th order low pass butter-worth filter.
High Pass Filter	: 3.4 KHz cut off.
Band Pass Filter	: Freq. range 7 to 13 KHz.
Band Reject Filter	: Freq. range 7 to 13 KHz.
Band Reject Filter	: Π Section: 20 KHz cut off. T Section: 20 KHz cut off.
Input Voltage	: 0 to 2 Vpp
Output Voltage	: 0 to 2 Vpp
Switch Fault	: 4 Switch faults are provided on board To study different effects on circuit
Interconnection	: 2mm banana socket.
Power Supply	: GND, +5, +12V, -12V.

Accessories:

ACL-FIL			Quantity
1	Mini Links PCS2 Patch cord	Male To Male	3
2	Short Links PCS2 Patch cord	Male To Male	2
3	Jumper Caps Small (5 mm)		5
4	Speaker Box Type		1
5	ACL-FIL EXPT Manual		1
6	Power Supply		1
7	Power cord		1

FUNCTIONAL BLOCK



INTRODUCTION

1.1 Analog Buffer

This is a unity gain amplifier built around op-amp IC 741 (U7)

1.2 Noise Generator

The noise source consists in a zener diode which, when operating in correspondence of the Breakdown zone, produces a white noise in the order of some tens of mV. The noise is amplified by two amplifier stages (U1 a/b) and applied to the adder stage U10A with a potentiometer for level regulation. The useful signal is applied, instead, to the amplifier U1c, which supplies it to the adder stage U1d. The gain of U9A varies from 0 to a maximum of 1, and, so, we can say that this stage attenuates the input signal. The output adder stage U1d adds the noise to the signal.

1.3 Audio Amplifier

In this circuit 0.5W amplifier LM 386 is used to amplify audio signal with total gain of 46dB. Volume can be controlled with the help of VOLUME pot. The output is obtained from the 8 ohm speaker when connected to the SPEAKER jack provided.

1.4 High Pass Filter

This filter is built around op-amp U1 (CA 3140) with cut off frequency of 3.4KHz. The filter is designed using following formula

$$F_L = 1 / [2 \pi (R1 C1 R2 C2)^{1/2}]$$

1.5 Band Pass Filter

This filter is built around op-amp U2 (CA 3140) with pass band frequency range of 7 KHz to 13 KHz and cut off frequency of 10 KHz. The filter is designed using following formula

$$F_H = 1 / [2 \pi (R1 C1 R2 C2)^{1/2}]$$

1.6 Band Reject Filter

This filter is built around op-amp U4 (CA 3140) with pass band frequency range of 7 KHz to 13 KHz and cut off frequency of 10 KHz. The filter is designed using following formula

$$F_L = 1 / [2 \pi (R1 C1 R2 C2)^{1/2}]$$

1.7 Matched Filter (T-Section)

This filter is built using the inductance L and capacitor C with cut off frequency of 20 KHz by using the following formula

$$F_C = 1 / [2 \pi (LC)^{1/2}]$$

1.8 Matched Filter (Π Section)

This filter is built using the inductance L and capacitor C with cut off frequency of 20 KHz by using the following formula

$$F_c = 1 / [2 \Pi (LC)^{1/2}]$$

1.9 Amplifier

This section provides amplification of analog signals with help of U8 (TL082) and the gain of the amplifier is fixed.

SWITCH FAULTS

Switch fault section on **ACL-FIL** is provided with **4** switches. These switches can be used to simulate fault conditions in various parts of the circuit.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. The bandwidth of band reject filter changes and it operates unevenly.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. The operation of the band pass filter fails and center frequency changes from 10 KHz to 14 KHz.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. The operation of the high pass filter fails and cut off frequency changes from 3.4 KHz to 14 KHz.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. The operation of the matched Π filter fails and cut off frequency changes from 20 KHz to 25 KHz.

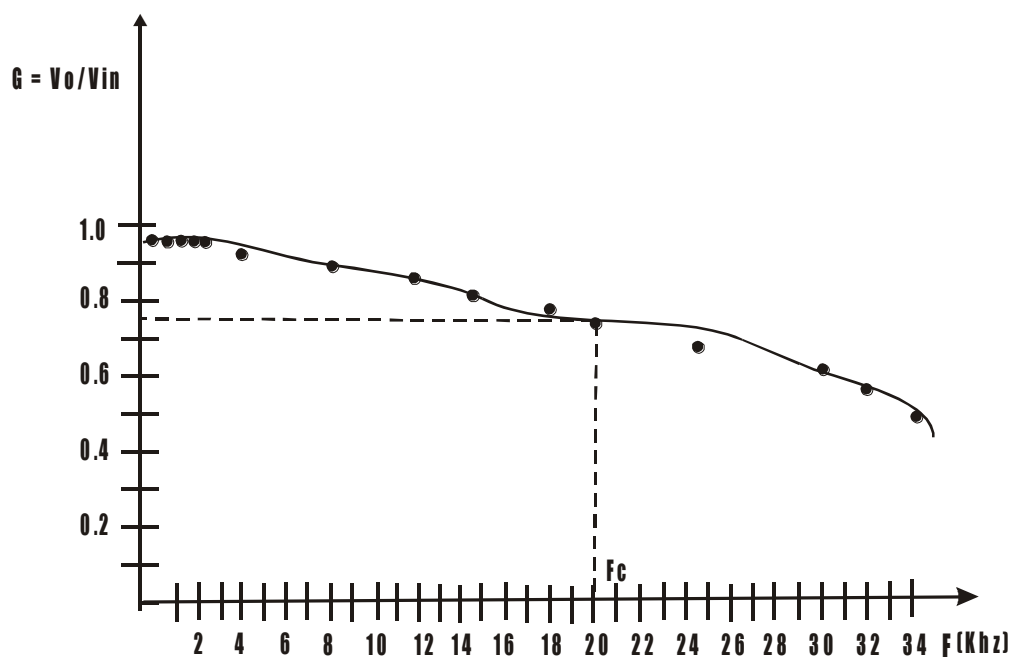
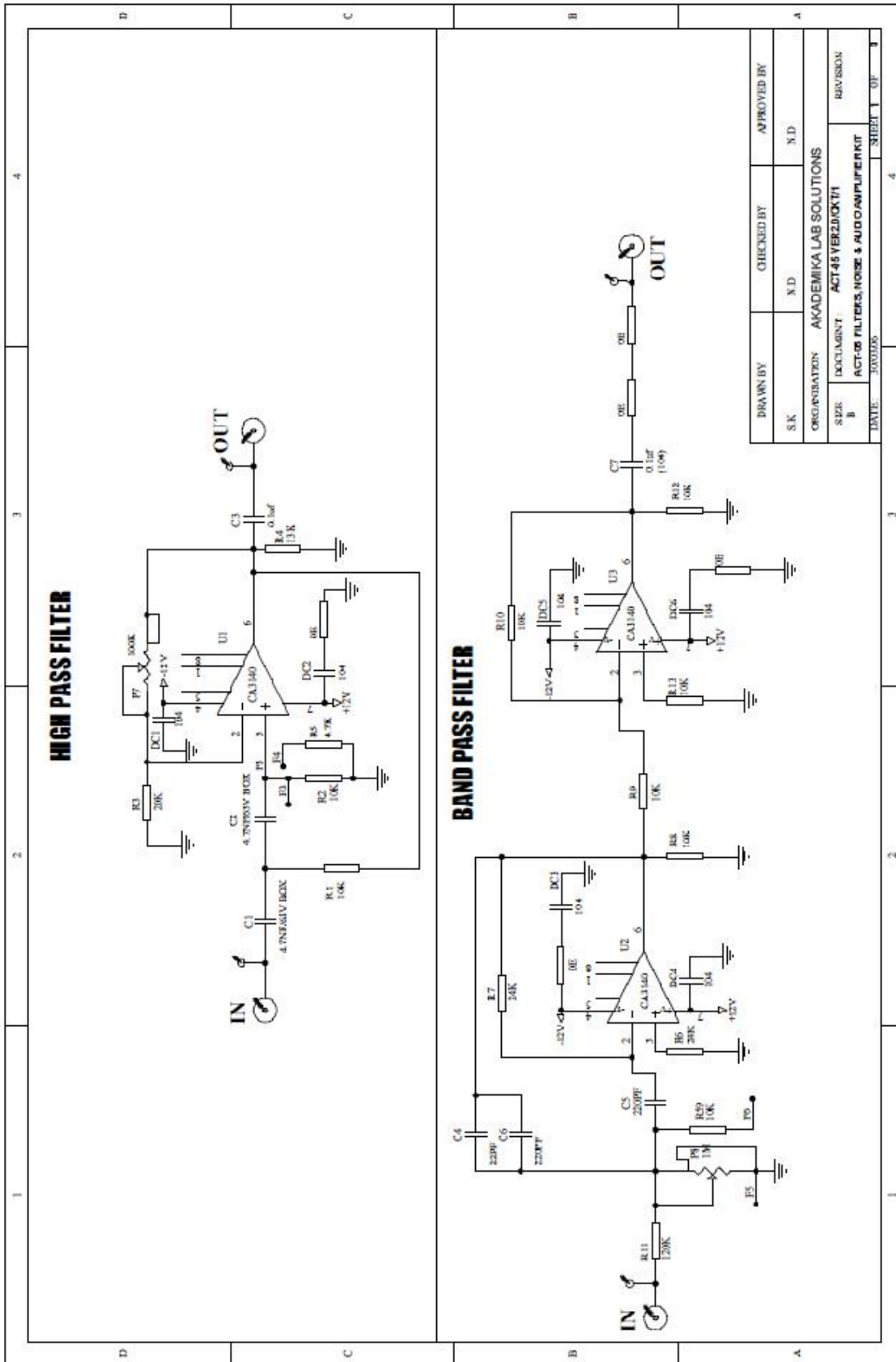
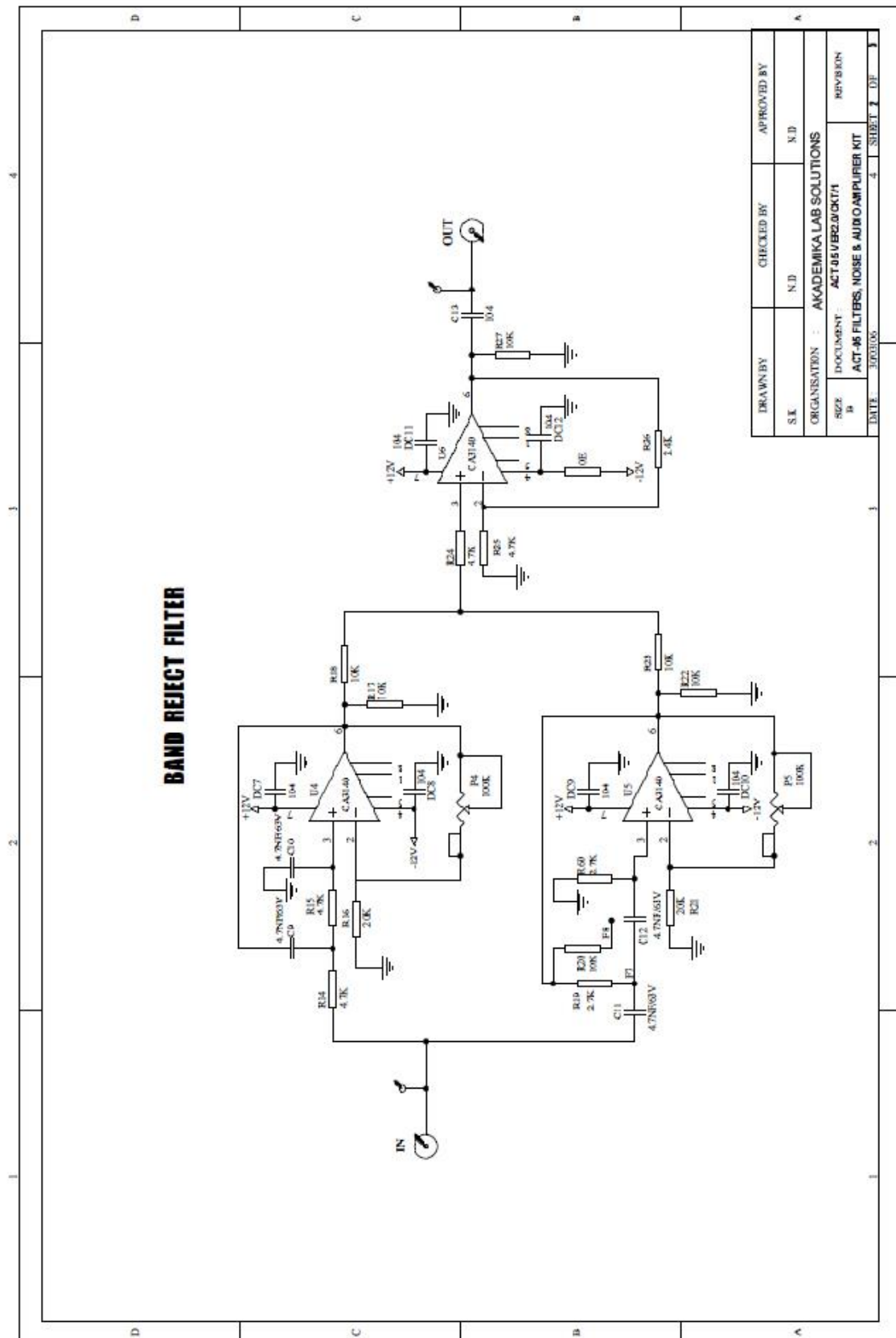
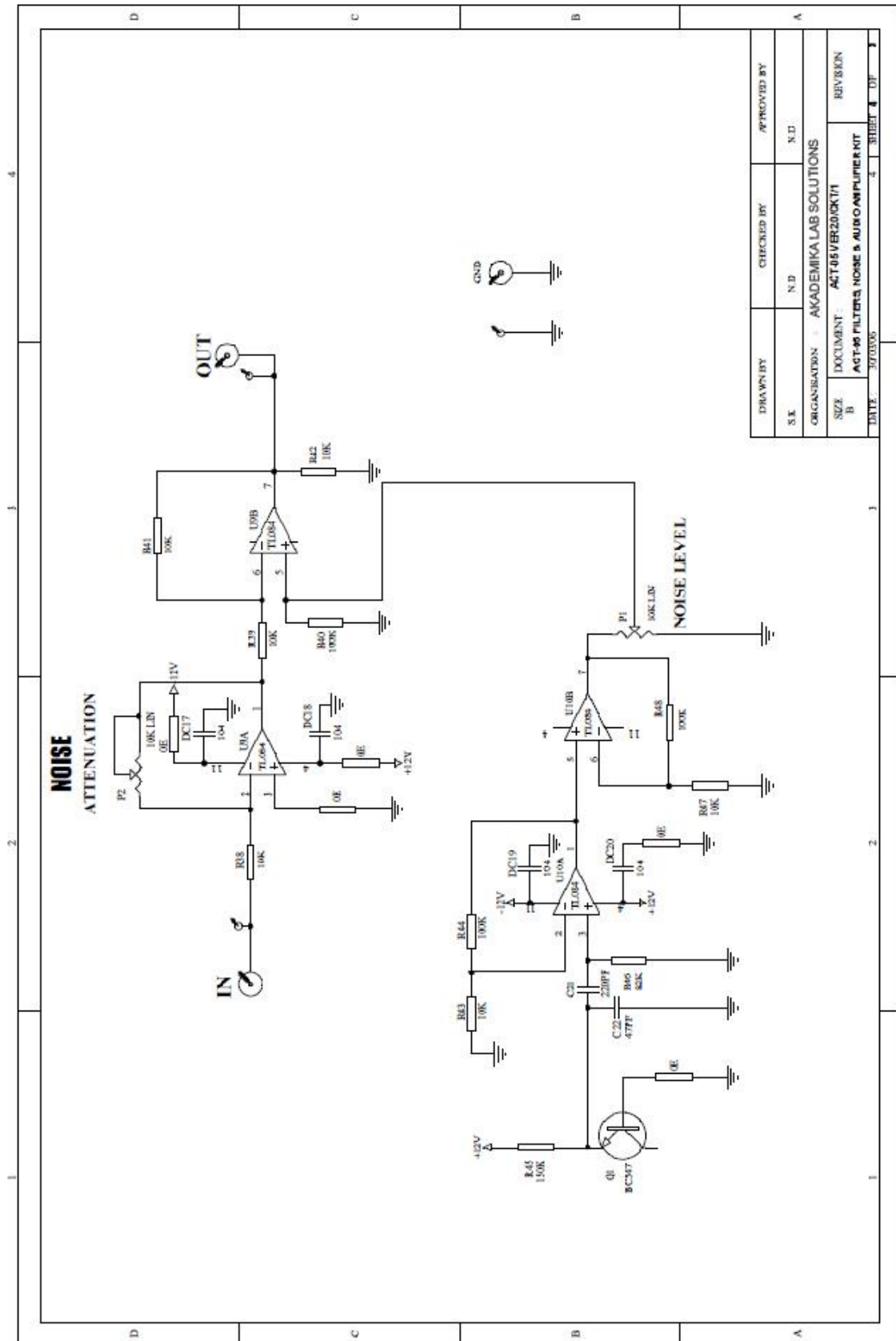


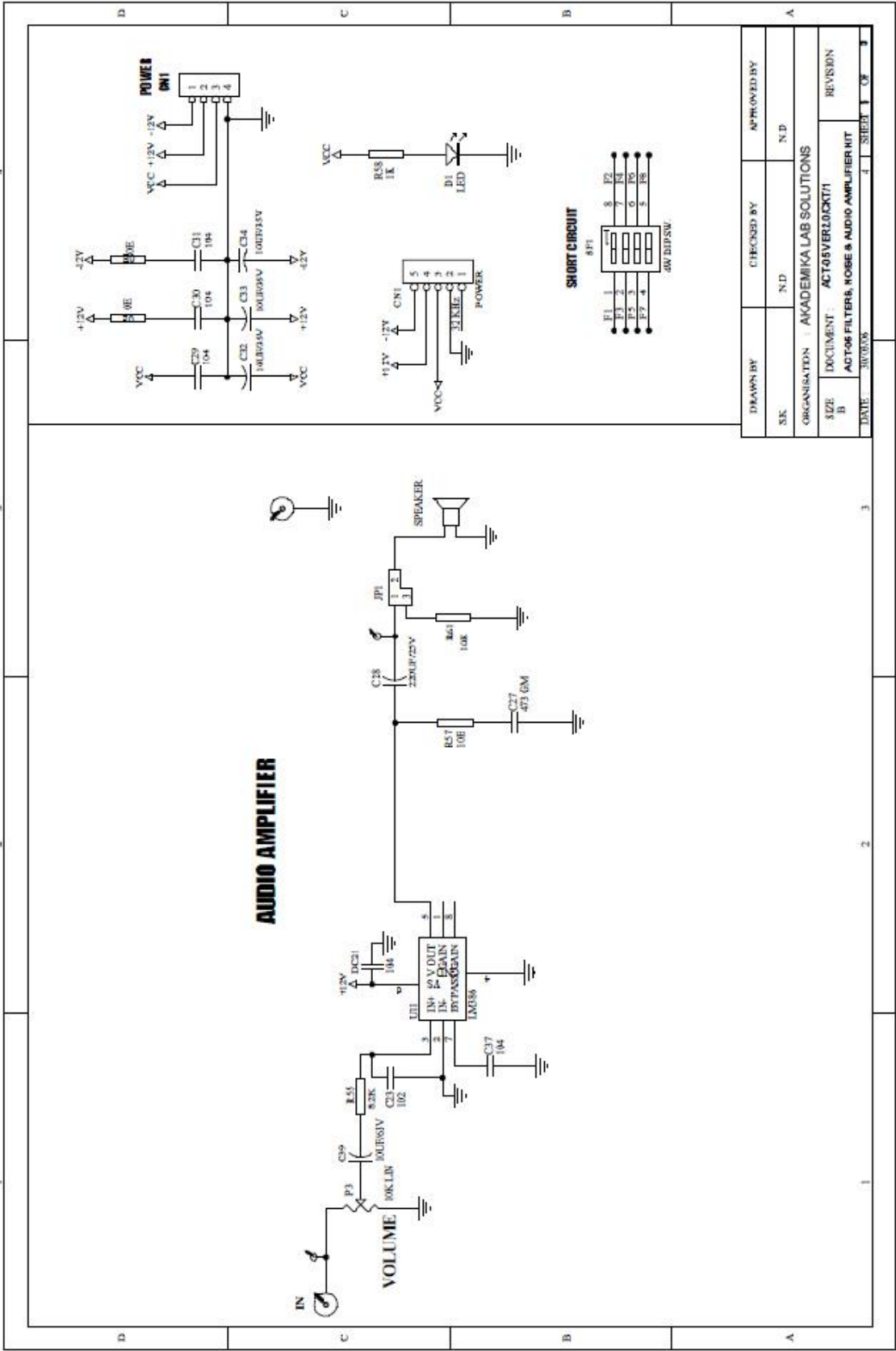
Fig. 3.14 FREQUENCY RESPONSE FOR MATCHED Π FILTER

CIRCUIT DIAGRAM

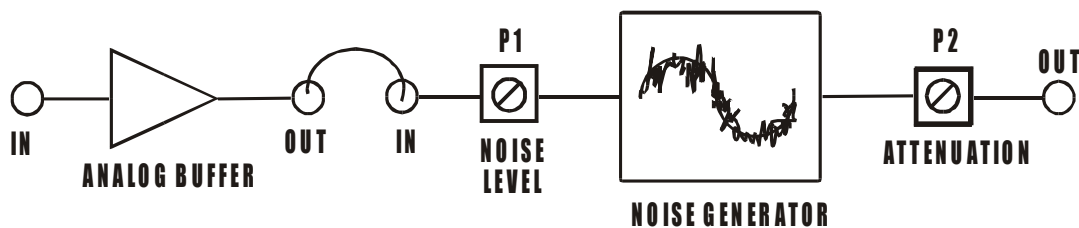
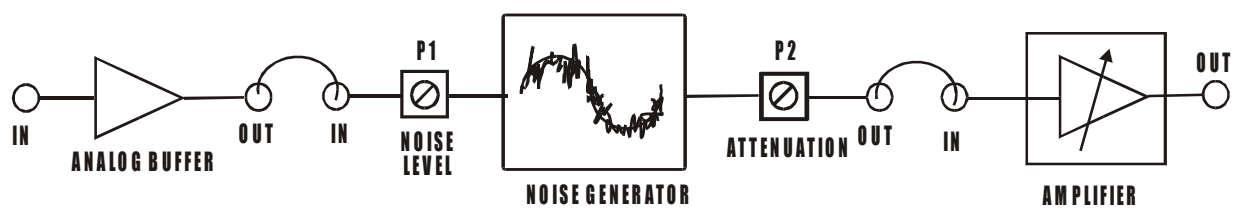








EXPERIMENT NO.1

**FIG.No 1.1 CHANNEL AND NOISE****FIG.No 1.2 MEASUREMENT OF NOISE FIGURE.**

EXPERIMENT NO.1

NAME:-

Study of Channel & Noise

OBJECTIVES:-

- A) Examine the operation of a noise generator.
- B) Examine the operation of a signal attenuation network.
- C) Measurement of S/N ratio.
- D) Measurement of Noise Figure.

THEORY:-

We are all familiar with the Noise and its ill effects not only on human health but also on electronic equipment. Apparently, 'Noise' is an undesirable electrical disturbance in a useful frequency band.

There are several sources of noise encountered in electronics and in everyday life. Some of the noise sources are: Howls, hum, ambient noise, back-ground noise, contact noise, circuit noise, component noise, carrier-wave noise, cosmic noise, flicker noise, line noise, modulation noise, random noise, radio receiver noise, shot noise, valve noise, transistor noise, thermal noise, diode noise, diode noise, pink noise, and white noise.

In electronic circuits (AF or RF) noise in the first and second stage must be kept extremely low otherwise the signal to noise ratio will become high and the incoming signal will be embedded in the pool of noise. We try to reduce noise by various means but the residual noise can't be removed fully.

So noise is not welcomed by circuit designers as also by the human beings. It is in fact a social embarrassment. There is a positive side to noise.

Noise source has become a very useful item in electronics. Noise source is used for adjusting VHF & UHF radio receivers. It is also essential for the synthesis of many sounds. These range from musical instruments such as pipe organ and percussive devices. We can now synthesize natural sounds like wind, rain and surf. In the field of man-made sounds such as steam engines, explosions and gunfire can be easily synthesized by noise.

What Is Noise?

Noise is caused by highly complicated physical and thermodynamic processes. Briefly, it is the random movement of electrical charge carriers. Noise increases with rise in temperature. At absolute zero ($-273^{\circ}\text{C}=0\text{K}^{\circ}$) noise is zero because at this temperature all movements of charge carriers are frozen, and there is no trace of noise.

Generation of Noise

The generation of noise was formerly very expensive but now it has become quite cheap thanks to the semiconductor technology which gave us the Zener diode and the transistor. These devices can be used at home industry to generate wide-band noise economically.

By reverse biasing a Zener diode above its break-down voltage, we can generate a lot of useful noise having a very wide frequency range from audio to R.F. (800 to 1000MHz). This wide band noise generated by the Zener diode is called White Noise.

The white noise can go far into radio frequencies therefore, white noise generators are used to inject noise frequencies into RF (radio frequency) amplifiers, communication radio receivers, and antenna bridges to evaluate small signal performance.

The white noise generators commercially available are very expensive test instruments which are beyond the pocket of amateurs and hobbyists. If anybody desires to have a white noise source, he can easily build a noise generator home using common components lying around him. The home built noise source can give usable results at much lower cost.

The Circuit

Refer to the ckt diagram. The noise source consists of a zener diode which, when operating in correspondence of the Breakdown zone, produces a white noise in the order of some tens of mV. The noise is amplified by two amplifier stages (U1 a/b) and applied to the adder stage U1d with a potentiometer for level regulation. The useful signal is applied, instead, to the amplifier U1c, which supplies it to the adder stage U1d. The gain of U1c varies from 0 to a maximum of 1, and, so, we can say that this stage attenuates the input signal. The output adder stage U1d adds the noise to the signal.

Signal to Noise Ratio

The calculation of the equivalent noise resistance of an amplifier, receiver or device may have one of two purposes or sometimes both. The first purpose is comparison of two kinds of equipment in evaluating their performance. The second is comparison of noise and signal at the same point. Therefore same point to ensure that the noise is not excessive In the second instance, and also when equivalent noise resistance is difficult to obtain, the signal-to-noise ratio S/N is very often used. It is defined as the ratio of signal power to noise power at

$$S/N = (V_s / V_n)^2$$

S = signal power & N = noise power.

Noise Figure

The noise figure F is defined as the ratio of the signal-to-noise power at the input of the amplifier to the signal-to-noise power at the output of the amplifier.

$$F = (SNR)_{i/p} / (SNR)_{o/p}.$$

EQUIPMENTS:-

- ACL-FIL
- Power supply
- 20MHz oscilloscope.
- Function generator

NOTE: Keep all switch faults in off condition

PROCEDURE:-

A) Examine the Operation Of A Noise Generator

- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 1.1 & carry out the following connections.
- Give the external sine wave signal of about **1KHz, 1Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of noise generator.
- Keep the noise level pot **P1** and attenuation pot **P2** in minimum position.
- Observe the signal at **OUT** post of noise generator.
- Increase the level of noise by varying the noise level pot **P1** of noise generator.
- Observe the signal at **OUT** post of noise generator. You will observe that the noise get superimposed on the original signal

B) Examine the Operation of a Attenuation Network

- Repeat the procedure from step no. 1 to step no.6.
- Keep the attenuation at maximum position by varying pot **P2**.
- Observe the signal at **OUT** post of noise generator and check that the output varies from a minimum of 0 to a maximum, which is about the input amplitude.
- Vary the input signal frequency and check that the channel presents an almost flat response.

C) Calculation of Signal To Noise (S/N) Ratio

- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 1.1 & carry out the following connections.
- Give the external signal **1KHz** about **1Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of noise generator.
- Keep the attenuation pot **P2** to maximum position i.e. completely clockwise direction.
- Keep the noise level pot **P1** of noise generator to get noise of 1Vpp amplitude.
- Measure voltage **Vn** at **OUT** post of noise generator.
- Keep the attenuation pot **P2** at minimum position i.e. completely anticlockwise direction.
- Measure voltage **Vs** at **OUT** post noise generator.
- Take further readings by increasing noise signal in steps of 0.5V as per step no.5 to step no.9.
- Calculate signal to noise ratio by using formula $S/N = (Vs / Vn)^2$

Noise amplitude Vn (volts)	Signal amplitude with noise Vs(volts)	(S/N)
1.0	3.3	10.89
1.5	3.8	6.41
2.0	4.6	5.29
2.5	5.8	5.38
3.0	6.3	4.41
3.5	7.2	4.23
4.0	8.2	4.20

(Table given for reference only actual values may differ from above)

D) Calculation of Noise Figure

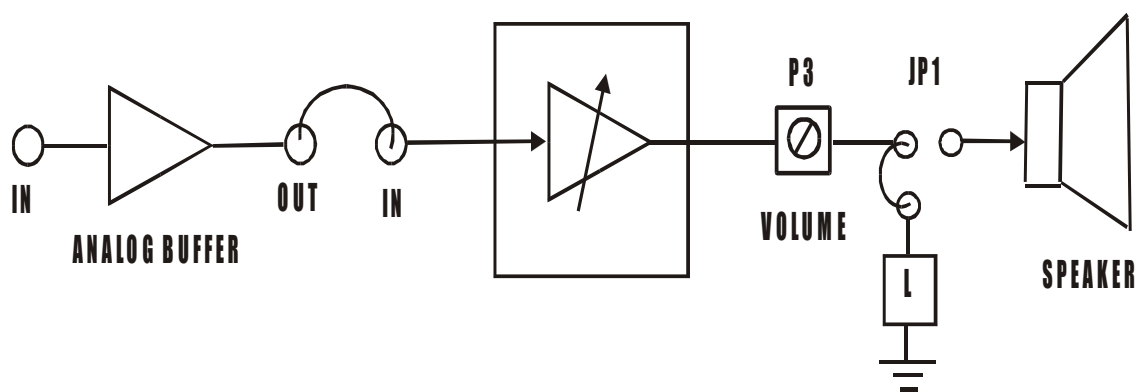
- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 1.2 & carry out the following connections.
- Give the external sine wave signal of about **1 KHz, 1Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of noise generator.
- Connect the **OUT** post of noise generator to **IN** post of amplifier.
- Keep the attenuation pot **P2** to maximum position i.e. completely clockwise direction.
- Keep the noise level pot **P1** of noise generator to get noise of 1Vpp amplitude.
- Measure voltage **Vni** at **IN** post of amplifier and **Vno** at **OUT** post of amplifier.
- Keep the attenuation pot **P2** to minimum position i.e. completely anticlockwise direction.
- Measure voltage **Vsi** at **IN** post of amplifier and **Vso** at **OUT** post of amplifier.
- Calculate signal to noise ratio by using formula $(\text{SNR})_{i/p} = (V_{si} / V_{ni})^2$
- Calculate signal to noise ratio by using formula $(\text{SNR})_{o/p} = (V_{so} / V_{no})^2$
- Take further readings by increasing noise signal in steps of 0.5V as per step no.6 to step no.12.
- Noise figure can be calculated by using following formula

$$\text{NOISE FIGURE} = (\text{S/N})_{i/p} / (\text{S/N})_{o/p}$$

Noise amplitude V_{ni} (volts)	Signal amplitude with noise at i/p V_{si} (volts)	$(\text{S/N})_{i/p}$	Signal amplitude with o/p V_{so} (volts)	Noise amplitude o/p V_{no} (volts)	$(\text{S/N})_{o/p}$	Noise figure
1	3.3	10.89	6.24	2	9.73	1.11
1.5	3.8	6.41	8.16	3	7.39	0.86
2	4.6	5.29	9.2	4	5.29	1.00
2.5	5.1	5.38	11.0	5	4.84	1.11
3.0	6.3	4.41	12.8	6	4.55	0.96
3.5	7.2	4.23	14.2	7	4.11	1.02
4.0	8.2	4.20	15.8	8	3.90	1.07

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.2

**FIG.2.1 AUDIO AMPLIFIER**

EXPERIMENT NO.2

NAME:-

Study of Audio Amplifier

OBJECTIVES:-

Measurement of the Frequency Response and Power output

THEORY:-

Refer to the circuit diagram as shown. In this circuit 6W amplifier TDA1010A is used to amplify audio signal with total gain of 54dB. Volume can be controlled with the help of VOLUME pot. The output is obtained from the 8 ohm speaker when connected to the SPEAKER jack provided.

EQUIPMENTS:-

- ACL-FIL
- Power supply.
- Low frequency generator
- 20MHz Oscilloscope.

NOTE: Keep All Switch Faults In Off Position

PROCEDURE:-

- Refer to the fig.2.1 & Carry out the following connections.
- Connect the power supply with proper polarity to the kit, while connecting this ensure that the power supply is OFF.
- Apply a sine signal with amplitude of 2Vpp at **IN** post of analog buffer.
- Connect OUT post of analog buffer to IN post of audio amplifier.
- Connect jumper **JP1**, connecting in this way the output of the amplifier to the load RL (10 Ω -resistor).
- Set the volume **P3** such as audio amplifier will have gain equal to 1 i.e. for 2V input we will get 2V output.
- Vary the signal frequency from 100 Hz onwards and detect the output signal amplitude at **OUT** post of audio amplifier.
- The gain of amplifier, measured at different frequencies, is given by:

$$G = V_o/V_i$$

And, in decibel:

$$G_{dB} = 20 \cdot \log (V_o/V_i)$$

The gain representation as function of the frequency determines the response curve of the amplifier.

- Calculate G and G_{dB} at each frequency and fill a table with the value of f and the corresponding values of V_o , G , G_{dB} . Plot a graph " $G_{dB} - f$ " with the values of f (in the x-axis) and the corresponding values of G_{dB} (in y-axis) ; you obtain a curve as per Datasheet of TDA1010A.
- Considering a reference gain at 1 KHz, evaluate the band width of amplifier. It is given by " $f_2 - f_1$ ", with f_1 and f_2 as frequencies at which the gain drops of 3dB in respect to the reference one.

Power Output

- Turn the volume to the maximum and increase the input signal (1-KHz frequency) until the output present no distortions.
- Calculate the power P_o supplied by the amplifier to the load R_L applying the formula:

$$P_o = (V_{eff})^2 / R_L \text{ with } V_{eff} = V_{opp} / 2\sqrt{2}; R_L = 10\Omega$$

F (KHz)	V_o (Vpp)	G	$G_{dB} = 20 \cdot \log (V_o/V_i)$
0.5	2	1	0
1.0	2	1	0
2	2	1	0
4	2	1	0
6	1.98	0.99	-0.08
8	1.96	0.98	-0.17
10	1.88	0.94	-0.53
12	1.84	0.92	-0.72
14	1.72	0.86	-1.31
16	1.68	0.84	-1.51
18	1.64	0.82	-1.72
20	1.52	0.76	-2.36

(Table given for reference only actual values may differ from above)

EXPERIMENT NO.3

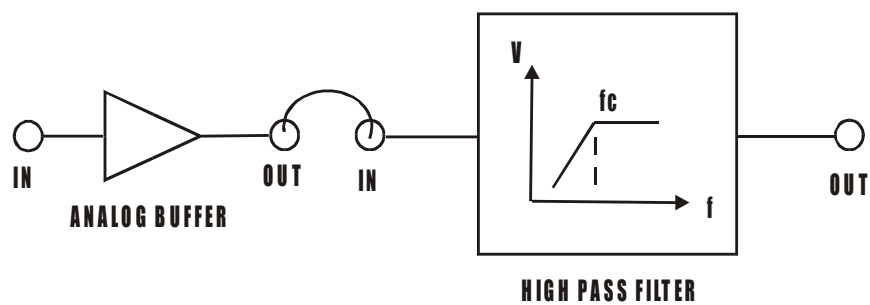


FIG. 3.1 STUDY OF FILTERS

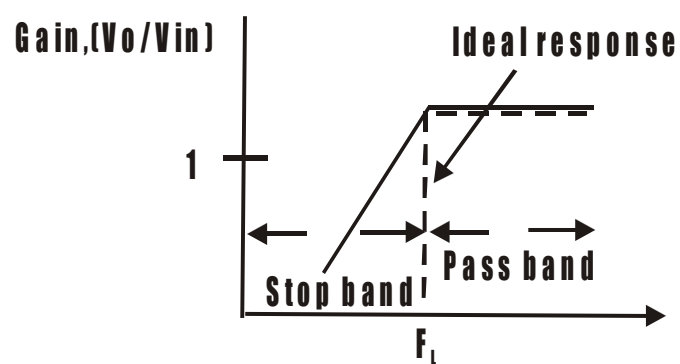


FIG. 3.2 FREQUENCY RESPONSE OF HIGH PASS FILTER

EXPERIMENT NO.3

NAME:-

Study of Filters

OBJECTIVES:-

- A) High pass filter.
- B) Band pass filter.
- C) Band reject filter.
- D) Matched filter (Π & T Section)

THEORY:-

High Pass Filters

High pass filters are often formed simply by interchanging frequency –determining resistors and capacitors in low pass filters. That is, a first order high pass filter is formed from a first order low pass type by interchanging components R and C. Similarly, a second order high pass filter is obtained from a second order low pass filter if R and C are interchanged.

Band Pass Filter

A band pass filter has a pass band between two cut off frequencies f_H and f_L such that $f_H > f_L$. Any input frequency outside this pass-band is attenuated. Basically there are two types of band pass filters.(1) wide band pass, and (2) narrow band pass. A filter is said to be wide band pass if its figure of merit or quality factor $Q < 10$. If $Q > 10$ then it will be narrow band pass filter. Thus Q is a measure of selectivity, meaning the higher the value of Q, the more selective is the filter or the narrower its bandwidth (BW). The relationship between Q, the 3-db bandwidth, and the center frequency f_c is given by

$$Q = f_c / BW = f_c / (f_H - f_L)$$

For the wide band pass filter the center frequency f_c can be defined as

$$f_c = (f_H f_L)^{1/2}$$

where f_H = high cut off frequency(Hz)

f_L = low cut off frequency of the wide band pass filter (Hz)

Wide Band Pass Filter

A wide band pass filter can be formed by simply cascading high pass and low pass sections and is generally the choice for simplicity of design and performance.

Band Reject Filter

The band reject filter is also called a band stop or band elimination filter. In this filter, frequencies are attenuated in the stop band while they are passed outside this band. The

band reject filter can also be classified as (1) wide band reject (2) narrow band reject. The narrow band reject filter is commonly called as notched filter. Because of its higher Q (> 10), the bandwidth of the narrow band reject filter is much smaller than that of the wide band reject filter.

Wide Band Reject Filter

A wide band reject filter is formed using a low pass filter, a high pass filter, and a summing amplifier. To realize a band reject response, the low cut off frequency f_L of the high pass filter must be larger than the high cut off frequency f_H of the low pass filter. The pass band gain of both the high pass and low pass sections must be equal.

Matched Filter

K-constant filters

The simplest structure of K-constant filter is shown in figure 3.1. The figure shows an example of pass band filter, the only one being examined as the results may be related to high pass, pass-band, notch filter.

The following relationship can be worked out for the circuit shown in fig. 4.1 by the theory of the passive quadruple networks.

$$Z_i = (L/C)^{1/2} \times (1-W^2LC)^{1/2} = Z_o (1-W^2LC)^{1/2} \quad \text{INPUT IMPEDANCE.}$$

$$Z_o = (L/C)^{1/2} \times 1/(1-W^2LC)^{1/2} = Z_o [1/(1-W^2LC)^{1/2}] \quad \text{OUTPUT IMPEDANCE}$$

$$A_i = \ln \left| (1-W^2LC)^{1/2} + jw(LC)^{1/2} \right| \quad \text{CONSTANT IMAGE ATTENUATION}$$

$$\text{FOR: } 0 \leq W \leq 1/(LC) (= W_c)$$

A_i takes 0 value, i.e. the network doesn't any attenuation. For $W > W_c$ A_i becomes:

$$A_i = \ln (w^2 L C - 1)$$

Under these conditions A_i is higher than zero and so the network introduces attenuation. Due to the different circuit behavior at different frequencies, this network may be used as a filter.

$$f_c = w_c/(2\pi) = 1/[2\pi(LC)^{1/2}]$$

is called filter cut off frequency.

The described network is called Constant-K because the product of the two impedances that are used (jwL and $1/jwC$) is constant when frequencies vary:

The change from low pass filter to high pass one is simply made by changing L with C .

When two "L" K-constant sections are put in cascade so as to stick the impedance match between the two sections themselves, the result is "Π" or "T" structure. When the cascade sections number grows, also the slope in the filter response growth

In practical applications, the filter source and load impedances usually resistive and constant like. As the input and output impedances of the K-constant filters, even though they are real inside the pass band they vary according to frequency, an impedance match can be obtained at a single frequency.

EQUIPMENTS:-

- ACL-FIL
- POWER SUPPLY.
- Low frequency generator
- 20MHz Oscilloscope.

NOTE: Keep All Switch Faults In Off Position

PROCEDURE:-

A) Calculate The Frequency Response Of The High Pass Filter

- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 3.1 & carry out the following connections.
- Give the external sine wave signal of about **1KHz, 2Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of high pass filter.
- Connect the oscilloscope with the filter output and list in a table the measured amplitude.
- Starting from the 1KHz, increase the generator frequency leaving the amplitude unchanged. List on the table the deviations in db of the output amplitude from the reference.
- Indicate in a graphic the table values.
- Calculate the filter bandwidth at 3db.

SWITCH FAULTS:-

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

Put switch **3** of **SF1** in Switch Fault section to **ON** position. The operation of the high pass filter fails and cut off frequency changes from 3.4KHz to 14KHz.

F (Hz)	Output Voltage Vo [Vpp]	Vo/Vi	20.log (Vo/Vi)
100	0.01	0.005	-46.02
500	0.16	0.08	-21.93
1000	0.28	0.14	-17.07
1500	0.52	0.26	-11.70
2000	0.74	0.37	-8.63
2500	1.00	0.50	-6.02
3000	1.28	0.64	-3.87
3400	1.40	0.70	-3.09
4000	1.60	0.80	-1.93
4500	1.68	0.84	-1.51
5000	1.84	0.92	-0.72
5500	1.92	0.96	-0.35
6000	2.00	1.00	0
6500	2.08	1.04	0.34
7000	2.16	1.08	0.66
7500	2.16	1.08	0.66
8000	2.18	1.09	0.74
8500	2.20	1.1	0.82
9000	2.20	1.1	0.82
10000	2.30	1.15	1.21
11000	2.30	1.15	1.21
12000	2.36	1.18	1.43
13000	2.36	1.18	1.43
14000	2.36	1.18	1.43
15000	2.36	1.18	1.43

(Table given for reference only actual values may differ from above)

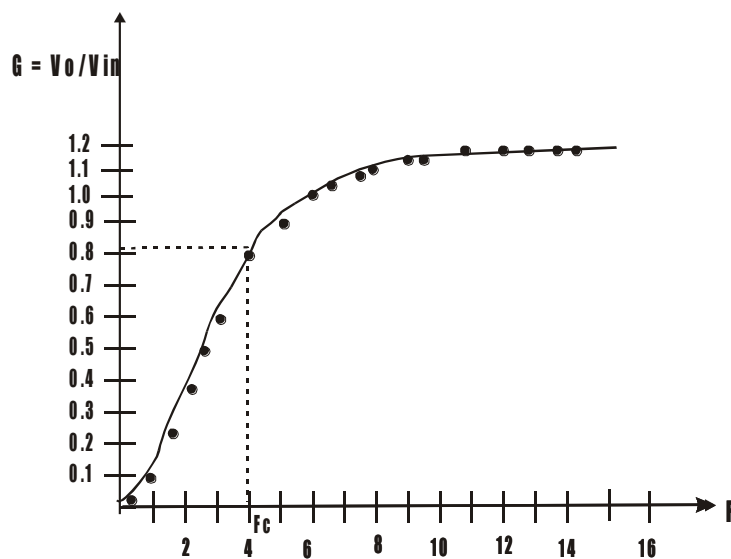


FIG.3.3 FREQUENCY RESPONSE FOR HIGH PASS FILTER

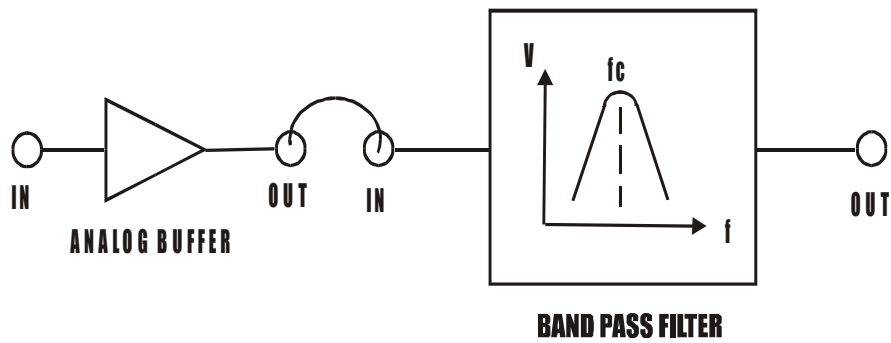


FIG.3.4 BAND PASS FILTER

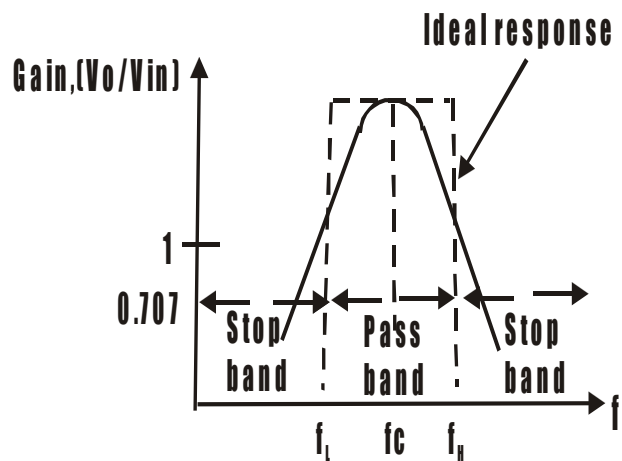


FIG.3.5 FREQUENCY RESPONSE OF BAND PASS FILTER.

B) Calculate the Frequency Response of The Band Pass Filter

- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 3.4 & carry out the following connections.
- Give the external sine wave signal of about **1KHz, 2Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of band pass filter.
- Connect the oscilloscope with the filter output and list in a table the measured amplitude.
- Starting from the 1KHz, increase the generator frequency leaving the amplitude unchanged. List on the table the deviations in db of the output amplitude from the reference.
- Indicate in a graphic the table values.
- Calculate the filter bandwidth at 3db.

SWITCH FAULTS:-

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

Put switch **2** of **SF1** in Switch Fault section to ON position. The operation of the band pass filter fails and center frequency changes from 10KHz to 14KHz.

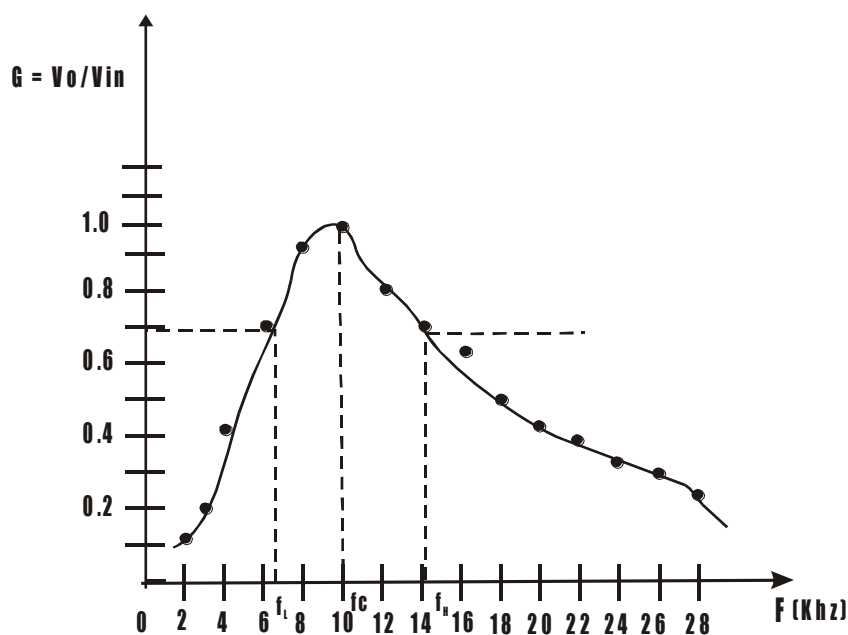


Fig. 3.6 FREQUENCY RESPONSE FOR BAND PASS FILTER

F (Hz)	Output Voltage Vo [Vpp]	Vo/Vi	20.log (Vo/Vi)
0.5	0.060	0.03	-30.45
1	0.120	0.06	-24.43
2	0.280	0.140	-17.07
3	0.520	0.260	-11.70
4	0.840	0.420	-7.53
5	1.16	0.580	-4.73
6	1.40	0.70	-3.09
6.5	1.44	0.701	-3.07
7	1.54	0.72	-2.85
8	1.86	0.93	-0.63
9	1.92	0.96	-0.35
10	2.00	1.00	0.00
11	1.84	0.92	-0.72
12	1.76	0.88	-1.11
13	1.56	0.78	-2.15
14	1.44	0.72	-2.85
15	1.36	0.68	-3.34
16	1.28	0.64	-3.87
17	1.16	0.58	-4.73
18	1.00	0.50	-6.02
19	0.926	0.46	-6.74
20	0.860	0.43	-7.33
21	0.820	0.410	-7.74
22	0.800	0.400	-7.95
23	0.760	0.380	-8.40
24	0.720	0.360	-8.87
25	0.640	0.320	-9.89
26	0.600	0.300	-10.45
27	0.560	0.280	-11.05
28	0.500	0.250	-12.04
29	0.460	0.230	-12.76
30	0.420	0.210	-13.55

(Table given for reference only actual values may differ from above)

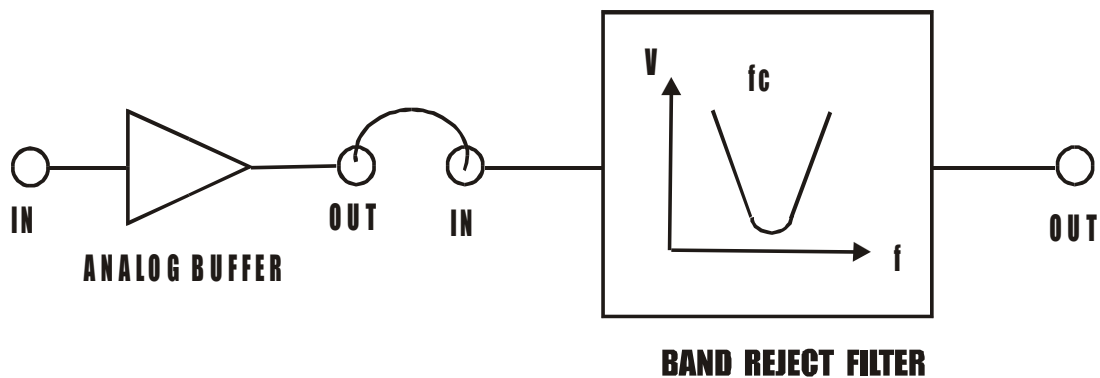


FIG.3.7 BAND REJECT FILTER

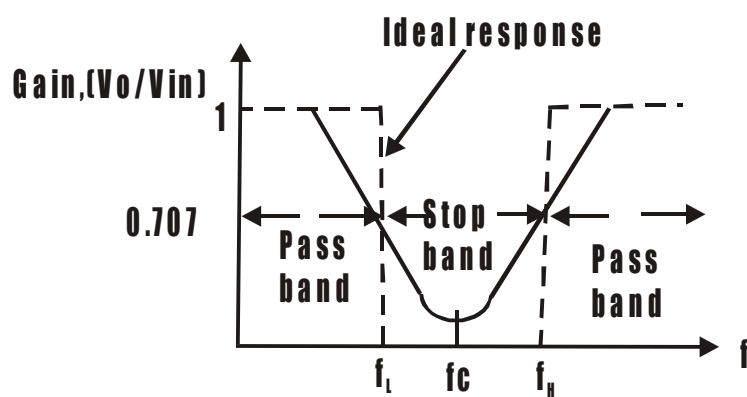


FIG.3.8 FREQUENCY RESPONSE OF BAND REJECT FILTER.

C) Calculate the Frequency Response Of Band Reject Filter

- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 3.7 & carry out the following connections.
- Give the external sine wave signal of about **1 KHz, 2Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of band reject filter.
- Connect the oscilloscope with the filter output and list in a table the measured amplitude.
- Starting from the 1KHz, increase the generator frequency leaving the amplitude unchanged. List on the table the deviations in db of the output amplitude from the reference.
- Indicate in a graphic the table values.
- Calculate the filter bandwidth at 3db.

SWITCH FAULTS:-

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

Put switch **1** of **SF1** in Switch Fault section to **ON** position. The bandwidth of band reject filter changes and it operates unevenly.

F (KHz)	Output Voltage Vo [Vpp]	Vo/Vi	20.log (Vo/Vi)
0.5	2.2	1.1	0.82
1.0	2.2	1.1	0.82
2.0	2.2	1.1	0.82
3	2.2	1.1	0.82
4	1.96	0.98	-0.17
5	1.92	0.96	-0.35
6	1.68	0.84	-1.51
7	1.50	0.75	-2.50
8	1.12	0.56	-5.03
9	1.00	0.50	-6.02
10	1.00	0.50	-6.02
11	1.20	0.60	-4.43
12	1.32	0.66	-3.60
13	1.50	0.75	-2.50
14	1.64	0.82	-1.72
15	1.76	0.88	-1.11
16	1.88	0.94	-0.53
17	1.92	0.96	-0.35
18	1.96	0.98	-0.17
20	2.12	1.06	0.51
22	2.20	1.10	1.28
23	2.24	1.12	0.98

(Table given for reference only actual values may differ from above)

D) Calculate the Frequency Response of Matched Filter

a) Matched Filter

- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 3.10 & carry out the following connections.
- Give the external sine wave signal of about **1KHz, 2Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of matched T filter.
- Connect the oscilloscope with the filter output and list in a table the measured amplitude.
- Starting from the 1KHz, increase the generator frequency leaving the amplitude unchanged. List on the table the deviations in db of the output amplitude from the reference.
- Indicate in a graphic the table values.
- Calculate the filter bandwidth at 3db.

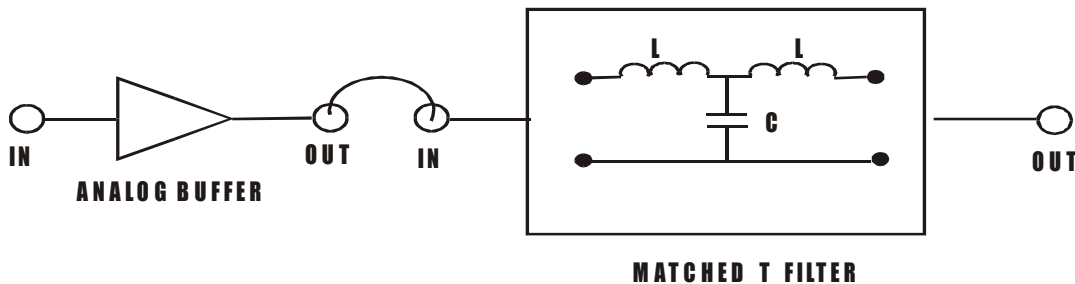


FIG. 3.10 MATCHED T FILTER

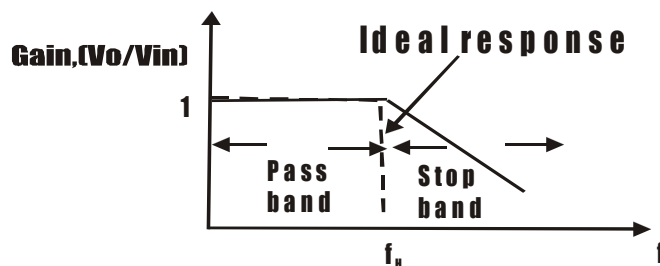


FIG. 3.11 FREQUENCY RESPONSE OF MATCHED FILTER.

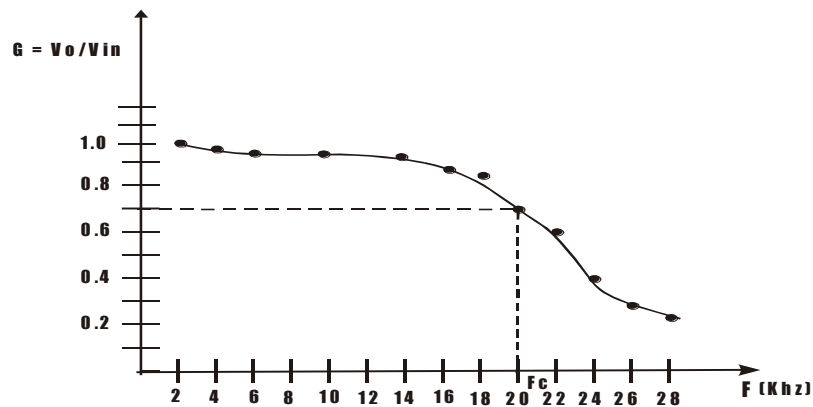


Fig. 3.12 FREQUENCY RESPONSE FOR MATCHED T FILTER

(Table given for reference only actual values may differ from below)

F (KHz)	Output Voltage Vo [Vpp]	Vo/Vi	20.log (Vo/Vi)
2	2	1	0
4	1.96	1.98	-0.17
6	1.92	0.96	-0.35
8	1.88	0.94	-0.53
10	1.86	0.93	-0.63
12	1.84	0.92	-0.72
14	1.82	0.91	-0.81
16	1.76	0.88	-1.11
18	1.68	0.84	-1.51
20	1.42	0.71	-2.97
22	1.20	0.60	-4.43
24	0.80	0.40	-7.95
26	0.560	0.28	-11.05
28	0.480	0.24	-12.39

b) Matched Π Filter

- Connect the power supply with proper polarity to the kit ACL-FIL and switch it **ON**.
- Ref to the fig. 3.13 & carry out the following connections.
- Give the external sine wave signal of about **1KHz, 2Vpp** to **IN** post of analog buffer.
- Connect the **OUT** post of analog buffer to **IN** post of matched T filter.
- Connect the oscilloscope with the filter output and list in a table the measured amplitude.
- Starting from the 1KHz, increase the generator frequency leaving the amplitude unchanged. List on the table the deviations in db of the output amplitude from the reference.
- Indicate in a graphic the table values.
- Calculate the filter bandwidth at 3db.

SWITCH FAULTS:-

NOTE: Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

Put switch **4** of **SF1** in Switch Fault section to ON position. The operation of the matched Π filter fails and cut off frequency changes from 20KHz to 25KHz.

F (Hz)	Output Voltage Vo [Vpp]	Vo/Vi	20.log (Vo/Vi)
100	1.92	0.96	-0.35
200	1.92	0.96	-0.35
500	1.92	0.96	-0.35
700	1.92	0.96	-0.35
1000	1.92	0.96	-0.35
1200	1.92	0.96	-0.35
1500	1.92	0.96	-0.35
1800	1.92	0.96	-0.35
2000	1.92	0.96	-0.35
4000	1.84	0.92	-0.72
8000	1.80	0.90	-0.91
12000	1.72	0.86	-1.31
15000	1.64	0.82	-1.72
18000	1.56	0.78	-2.15
20000	1.44	0.73	-2.73
25000	1.36	0.68	-3.34
27000	1.32	0.66	-3.60
30000	1.24	0.62	-4.15
50000	0.88	0.44	-7.13
100000	0.48	0.24	-12.39

(Table given for reference only actual values may differ from above)